Tutorial 4 –Experimental error (2)

Background

- How to calculate the mean, SD and RSD for a set of data in the measurement?

$$\bar{x} = \frac{\Sigma_i(x_i)}{N} s = \qquad \text{SD} = \sqrt{\frac{\Sigma_i(x_i - \bar{x})^2}{N-1}} \qquad RSD = \frac{s}{\bar{x}} \times 100\%$$

- How to report the experimental result with different interests?

Exercise

T4-1. A chemist performs analysis of chlorine in water, for each experiment, he performs 5 measurements. He repeats the experiment 3 times, as shown in the table:

Exp. 1	
No. #	**[Cl^-] ppm**
1	1.032
2	0.998
3	1.014
4	1.026
5	1.029
6	1.003

Exp. 2	
No. #	**[Cl^-] ppm**
1	0.990
2	0.970
3	0.993
4	0.967
5	0.968
6	1.014

Exp. 3	
No. #	**[Cl^-] ppm**
1	1.010
2	0.958
3	1.008
4	0.998
5	1.032
6	0.950

a. Calculate the means ($\bar{x}$) for each experiment.

b. Calculate the standard deviation (S) for each experiment.

c. Which experiment had the best precision for the three experiments?

T4-2. Find the relative standard deviation (RSD) for the numbers of 116.0, 97.9, 114.2, 106.8, and 108.3.

T4-3. The pressure in a system was measured many times. The following results were calculated:

$\bar{x}$ = 3.12356 Mpa

SD = 0.16789%

n = 87

Rewrite the results according to the presentation rules you have learned (keeping a correct number of significant digits), for the following two cases:

a. You are interested in the measuring method.

b. You are interested in the results.

T4-4. Two people entered a star observatory; the first person said that god told him in a dream that there are 336 stars in $1m^2$ of sky. The second person did not believe it, and wanted to check whether the first person is indeed connected to god. Therefore, two of them counted the number of stars in 1 m^2 of sky 5 times together. These are their results:

Count number	**Stars in 1 m^2 of sky**
1	335
2	337
3	330
4	337
5	320

a. In the above experiment what is μ and what is $\bar{x}$?

b. Calculate the standard deviation for each person for these results. What does standard deviation represent?

The second person repeated the count for another two days and got the following results:

The second day	
Count number	**Start in $1\ m^2$ of sky**
1	324
2	341
3	331
4	334
5	331

The third day	
Count number	**Start in $1\ m^2$ of sky**
1	340
2	327
3	333
4	335
5	345

c. Calculate the mean number of stars in 1 m^2 of sky for every day, and calculate the mean value of stars in 1 m^2 of sky for the whole 3 days.

d. Calculate the standard deviation for the means. What does it represent?

e. Assuming the first person is right, in which day the precision is the highest and in which day the accuracy is the highest?